Mission 6: The Recovery Clinic

The story

"Day 14. Arben is home, and he comes back to the clinic for a check. The doctor wants to know how well his brain is recovering, and the patient wants a number, not a feeling. Your sixth mission turns the terminal into three short brain tests, saves every visit, and compares each result with normal people of the same age."

What you need to know

- After a stroke, patients come back for visits, for example on day 14, day 45 and day 90 after the stroke.
- At each visit they do three short tests. Each test gives one number.
- A number means little alone. It is compared with the **normal average** for people of the same age, and with the **spread**: how far healthy people usually are from that average.

Test	What the patient does	Result	Better when
Reaction time	Presses ENTER as soon as a `*` appears	average milliseconds	lower
Memory span	Repeats a growing row of digits	longest row repeated correctly	higher
Colour words	Names the colour a word is printed in, not the word	seconds, with a penalty for mistakes	lower

The base challenge

Your goal: run three short brain tests in the terminal, keep every visit, and tell the patient how each result compares with healthy people of the same age.

Goal 1: Read the normal values

We give you `data/norms.txt`, in the same format as the other files. The values are made up for this competition.

```
# NORM | test | from age | to age | normal average | spread

NORM | reaction  | 0  | 29  | 300 | 40
NORM | reaction  | 30 | 49  | 320 | 45
NORM | reaction  | 50 | 64  | 350 | 50
NORM | reaction  | 65 | 79  | 390 | 60
NORM | reaction  | 80 | 120 | 430 | 70
NORM | memory    | 0  | 29  | 7.0 | 1.2
NORM | memory    | 30 | 49  | 6.8 | 1.2
NORM | memory    | 50 | 64  | 6.5 | 1.2
NORM | memory    | 65 | 79  | 6.0 | 1.1
NORM | memory    | 80 | 120 | 5.5 | 1.1
NORM | colorword | 0  | 29  | 14  | 3
NORM | colorword | 30 | 49  | 15  | 3
NORM | colorword | 50 | 64  | 16  | 3.5
NORM | colorword | 65 | 79  | 18  | 4
NORM | colorword | 80 | 120 | 21  | 5
```

Goal 2: Build the three tests

Each test is its own piece of code that returns one number.

Reaction time

1. 5 rounds. Each round prints `Wait...`, waits a random time between 1 and 3 seconds, then prints `*`.
2. Measure the milliseconds from printing `*` to the patient pressing ENTER.
3. A time under 100 ms means ENTER was pressed too early: print `Too soon!` and repeat the round.
4. Result: the average of the 5 rounds, in whole milliseconds.

Memory span

1. Show 3 random digits, for example `7 2 9`. No digit appears twice in a row.
2. Leave them on screen for 1 second per digit, then hide them so the patient cannot read them any more.
3. The patient types the digits back. Spaces do not matter.
4. Correct: one more digit next time. The first mistake ends the test. The longest row is 9 digits.
5. Result: the longest row repeated correctly (0 if the first row was wrong).

Colour words

1. 10 rounds. Each round prints one word, `RED`, `BLUE`, `GREEN` or `YELLOW`, in a colour. In at least 7 rounds the colour is different from the word.
2. The patient types the first letter of the **colour**, not of the word: `r`, `b`, `g` or `y`. Any other key: ask again, it does not count.
3. Result: the total seconds for the 10 rounds, plus 2 seconds for every wrong answer, with one decimal.

Goal 3: Run and save a visit

`clinic visit <case id>` asks how many days have passed since the stroke, runs the three tests one after another, and saves the visit in the case.

`clinic record <case id>` saves a visit from results typed in by hand, for tests done on paper. The judges use it to test your program with exact numbers.



Example CLI interaction (you may design yours differently):

```
synapse> clinic visit P-0001
Days since the stroke: 90

Test 1 of 3: reaction time
Press ENTER as soon as you see *   (5 rounds)
Wait...
*
412 ms
...
Average: 430 ms

Test 2 of 3: memory span
Remember the digits. They disappear after a moment.
...
Longest row repeated: 6 digits

Test 3 of 3: colour words
Type the first letter of the COLOUR of each word, not the word: r, b, g or y.
GREEN
> r
...
Time: 20.0 s, 1 wrong answer (+2 s) = 22.0 s

SUCCESS: Visit on day 90 saved for P-0001.

synapse> clinic record P-0001
Days since the stroke: 14
Reaction time (ms): 540
Memory span (digits): 4
Colour words (seconds): 31
SUCCESS: Visit on day 14 saved for P-0001.
```

In the terminal, the word `GREEN` above appears in red ink.

Example of the saved visits inside the case (you may design yours differently):

```
"visits": [
  { "day": 14, "reactionMs": 540, "memoryDigits": 4, "colorWordSeconds": 31.0 },
  { "day": 45, "reactionMs": 470, "memoryDigits": 5, "colorWordSeconds": 26.0 },
  { "day": 90, "reactionMs": 430, "memoryDigits": 6, "colorWordSeconds": 22.0 }
]
```

Goal 4: Compare with normal values

A result alone means little: a reaction time of 470 ms is normal at 80 and well below normal at 20. `clinic results <case id>` shows every visit, oldest first, and rates each result against the normal values for the patient's age. Count how many spreads the result is worse than the normal average:

- **normal**: up to 1 spread worse, or better than the average;
- **below normal**: more than 1 and up to 2 spreads worse;
- **well below normal**: more than 2 spreads worse.

Worse means slower for reaction time and colour words, and fewer digits for memory span.



```
synapse> clinic results P-0001
Arben Hoxha, 67 · normal for ages 65–79: reaction 390 ms, memory 6.0 digits, colour words 18.0 s

DAY  REACTION              MEMORY               COLOUR WORDS
14   540 ms  well below    4 digits  below      31.0 s  well below
45   470 ms  below         5 digits  normal     26.0 s  below
90   430 ms  normal        6 digits  normal     22.0 s  normal
```

Error handling

Your program must handle and clearly report errors such as:

- "ERROR: No case called "P-0099"."
- "ERROR: Days since the stroke must be a whole number of 1 or more."
- "ERROR: P-0001 already has a visit on day 14."
- "ERROR: Reaction time must be a whole number of milliseconds above 0."
- "ERROR: norms.txt, line 7: ages 60 to 70 overlap with line 6."
- "ERROR: No normal values for age 130."
- "Case P-0003 has no visits yet." (a message, not a crash)

The bonus challenge

For extra points.

Bonus 1: Trend

Compare the last two visits for every test:

- Reaction and colour words: **improving** if the new result is more than 5% lower, **worse** if it is more than 5% higher, otherwise **steady**.
- Memory: **improving** if higher, **worse** if lower, otherwise **steady**.

For Arben, day 90 against day 45, all three tests are improving.

Bonus 2: Does it match the damage?

Some weak results are expected, because the stroke damaged the areas behind them. A weak result that the damage does not explain is a warning. A test is expected to be weak if one of these areas stopped working:

Test	Expected to be weak if one of these areas stopped working
Reaction (pressed with the right hand)	`right-arm-face`, `awake`
Memory span	`memory`, `understanding`
Colour words	`understanding`, `left-attention`, `right-vision`, `left-vision`

`clinic check <case id>` raises a flag when a latest result is below normal but the damage does not explain it, or when a test is getting worse. Either one can mean a new stroke.



Drita (P-0002, age 72, blocked pipe `right-neck`) has three recorded visits:

Day	Reaction	Memory	Colour words
14	420 ms	6 digits	30 s
45	410 ms	6 digits	25 s
90	405 ms	3 digits	24 s

```
synapse> clinic check P-0002
Damage: right-neck blocked -> left-arm-face, left-attention, left-leg stopped working
TEST          DAY 90               TREND    EXPECTED WEAK
Reaction      405 ms  normal       steady   no
Memory        3 digits  well below worse    no
Colour words  24.0 s  below        steady   yes
FLAG: Memory is well below normal, and the damage does not explain it.
FLAG: Memory is getting worse since day 45.
Advice: check the patient for a new stroke.
```

Her colour words result is below normal, but it raises no flag: the damage to `left-attention` explains it.

Bonus 3: Chart in the terminal

`clinic chart <case id> <test>` draws one bar per visit: one `#` per 20 ms for reaction, one per digit for memory, one per second for colour words, rounded to the nearest whole `#`. Print the normal range (average ± 1 spread) above it.

```
synapse> clinic chart P-0001 reaction
Reaction time · lower is better · normal for age 67: 330 to 450 ms
Day 14  ############################  540 ms
Day 45  ########################      470 ms
Day 90  ######################        430 ms
```

Hints

- Not every clock in your language is precise enough for milliseconds. Research which one to use.
- Research how a terminal prints coloured text.
- The three tests are different, but the clinic runs them the same way, one after another. Think about what they have in common.
- Try the tests on yourselves. Tiredness and stress slow anyone down.

What the judges will test

- Visits typed in with `clinic record`, compared with the judges' answer sheet.
- Ages on the edges of the norm bands: 64 and 65, 79 and 80.
- Visits recorded out of order (day 45 before day 14): results are still shown oldest first.
- A case with one visit only: no trend, no crash.
- Each test played once by hand, including pressing ENTER too early.

What this mission saves

The visits inside the case. Mission 7 prints them at the end of the patient's journey.

